

SOLUTION ARCHITECTURE & DATA DESIGN

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Layered Application Design

- Presentation Layer: HTML pages, custom CSS glassmorphism, and Bootstrap 5 responsive elements.
- Controller Layer: Flask endpoints (app.py) that manage forms, invoke models, and log predictions.
- Model Pipeline Layer: Saved pickle (.pkl) files (scalers, encoders, estimators).
- Data Persistence: local CSV history tracker and SQLite database files.

Layer 1 - Client UI

2. Data Schema and Log Structure

HTML5 forms, CSS3 glassmorphism, Bootstrap 5 responsive grid

User prediction history is written to 'datasets/user_predictions.csv' with the following schema:

Layer 2 - Flask Server

Field Name		Data Type	Description
/predict POST route: validates inputs -> scales > infers > logs to CSV			
N / P / K	/dashboard GET route: renders analytics charts	Float	Nutrient levels (kg/ha).
temperature / humidity	Layer 3 - ML Pipeline	Float	Climatic values.
ph / rainfall	best_model.pkl (RF Classifier)	Float	Soil pH and rainfall (mm).
crop	yield_model.pkl (RF Regressor)	String	Recommended crop name.
yield	scaler.pkl (StandardScaler)	Float	Expected yield (t/ha).
soil_cluster	kmeans_model.pkl(K-Means, k=5)	Integer	K-Means category ID (0-4).
location / season	datasets/user_predictions.csv - append-only prediction log	String	Region and planting season.